Validation and Calibration of the Remote Consolidated Approach of Reporting on Food Security (rCARI) Methodology

Preliminary results from Burkina Faso pilot study

Version 1.0

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Executive Summary

The Remote Consolidated Approach for Reporting Food Insecurity (rCARI) is an adaptation of the standard CARI methodology, designed specifically for use with data collected through remote modalities. While rCARI retains the core food security indicators of the conventional CARI, it diverges in the measurement of household vulnerability. In place of Food Expenditure Share (FES) or Economic Capacity to Meet Essential Needs (ECMEN)—which are difficult to capture reliably through phone-based surveys—rCARI incorporates indicators related to income source and income change. Although rCARI is currently being piloted in various assessments, it has yet to undergo empirical validation. This study aims to validate the rCARI methodology by comparing its outcomes with those of the standard CARI using data from the same sample populations.

This study employed a survey experiment using a stratified sample to evaluate and validate the rCARI methodology against the standard CARI methodology. The sampling frame comprised the population from three purposively selected regions—Plateau Central, Centro-Nord, and Nord Regions (ADMIN1). These regions were chosen based on mobile phone coverage, food security status, and accessibility, considering on-ground security challenges. The study sample was randomly selected from a full household census across 38 villages (ADMIN5) or primary sampling units (PSUs). In total, 4,500 households were drawn randomly (using the random number approach) from the census list of 10,554 households.

Sampled households were then randomly assigned to one of two study groups (survey modalities):

1. **Face-to-Face (F2F) interviews:** Households received standard CARI modules.
2. **Remote (phone) interviews:** Households received rCARI modules.

In Burkina Faso, challenges such as poor infrastructure and limited accessibility affected field operations. For instance, two ADMIN5 areas were excluded from the study due to security and inaccessible road network. The study ultimately interviewed 2,882 households out of a potential 4,256 households, with 1,863 (65%) interviewed F2F and 1,019 (35%) interviewed remotely.

This report presents the preliminary analysis of data collected between 1–30 September 2024 in Burkina Faso and follows these steps:

- **Comparison of main food security indicators:** The main food security indicators (Food Consumption Score (FCS), Food Expenditure Score (FES), Reduced Coping Strategies Index (rCSI) and Livelihood Coping Strategies Index (LCSI)) are compared across F2F and remote modalities.
- **Calculation and validation of CARI Versions:**Three variations of CARI are calculated and compared against the standard CARI:
    1. CARI with FES that is collected in the remote modality (rCARI-FES).
    2. CARI with income source & income change (F2F version), (CARI-Inc).

3. CARI with income source & income change (remote version): This version was developed during the COVID-19 pandemic and is referred to as the standard rCARI in this report (rCARI-Inc).

The results highlight the following key findings:

- In estimating the prevalence of acute food insecurity (AFI) defined as the moderately plus severely food insecure groups, remote CARI using income source and income change indicator reported a significantly lower prevalence (approximately 15 percentage points on average) compared to standard CARI results from F2F modules. This difference was significant across all four CARI categories.
- Using the pilot shortened expenditure module implemented in the remote surveys produced results that did not significantly deviate from the standard CARI results in terms of the aggregated AFI.
- The results further show that the income choice and income change indicator remained robust, stable and consistent in under stating households' vulnerability regardless of the interview modality (F2F or remote). For example, respondents always underestimated their economic vulnerability when the income source and income change indicator was used by between 14 and 17 percentage points compared to when the full expenditure module (FES) regardless of survey modality.
- In general, comparative households tend to report better food security outcome indicators over the phone than those interviewed in F2F.This leads to better remote overall outcome than the F2F.
- Regression analysis, controlling for baseline respondent characteristics, showed no statistically significant effect of survey modality on food insecurity group (being classied as food insecure, moderately food insecure and severely food insecure combined), regardless of whether FES or the income source and income change indicator was used. However, the gender of the respondent was significantly associated with the probability of the household being food insecure across both methods (CARI-F2F and rCARI) suggesting potential gender-specific vulnerabilities or reporting biases.

The overarching communication points from the foregoing is that:

- **Survey Modality (F2F vs. Remote):** While challenges such as connectivity and lower response rates may affect remote surveys, when properly implemented they can produce food security assessments outcomes that are comparable to the standard F2F surveys results.
- **Alignment of CARI and rCARI Methodologies:** Compared to FES, the income source and income change indicator as a vulnerability measure tends to under report the severity and prevalence of AFI regardless of the survey modality.

1 Introduction

Food security assessment and reporting are at the core of WFP's mandate of addressing hunger and food insecurity. Over time, different methodologies have been developed (e.g, IPC, CARI, Cadre Harmonise) and used to assess and report global food insecurity each differing in scope, and purpose. A commonality that cuts across all methods is the use of primary household data collected through face-to-face interviews/surveys. However, in recent times particularly with the outbreak of the Covid-19 pandemic and attendant restrictions on movement and physical contact, where data collection through conventional face-to-face method has been adversely affected. This negatively impacted the ability to assess food security needs because of the data gap created. Developing methodologies to accurately measure and interpret food insecurity especially in emergencies where physical access is restricted thus, is of great importance for WFP and many stakeholders. This is particularly important during emergencies when rapid assessment is required for advocacy and policy to save lives and livelihoods but physical access to the field is restricted either because of health threats or security considerations for enumerators.

To leverage the increased penetration of technology in operational areas of WFP during the Covid lockdowns, a remote version of the Consolidated Approach for Reporting Food Insecurity (CARI)- the so-called rCARI- was developed as an adaptation of the standard CARI methodology and administered remotely (over the phone). The methodology provided a stop-gap solution during the lockdown, however, questions about the validity of food security numbers it produces remain. For example, how are the remote modules comparable with already standardized face-to-face modules in capturing food insecurity. Although there exist validated studies on the use of face-to-face modules in assessing and reporting on food insecurity, remote methodology such as rCARI remains scant until now.

The potential advantages of the methodology abound for humanitarian work particularly in hard-to-reach areas where F2F assessment is practicably impossible. For example, the remote methodology has potential to save lives by providing real-time food insecurity data on the exposed population within the shortest possible time to inform policy and advocacy purposes. Furthermore, it can result in better use of scarce resources to achieve high end results by rolling out few targeted modules to assess food insecurity of a larger population.

The overall objective of this study is to assess the comparability of the remote consolidated approach for reporting food insecurity - rCARI modules with the conventional CARI modules for assessing and reporting food insecurity. Specifically, the following indicators; Food Consumption Score (FCS), reduced Coping Strategy Index (rCSI), Livelihood Coping Strategy Index (LCSI), Food Expenditure Share (FES), and income sources and income change have been compared over the two modalities. The Economic Capacity to Meet Essential Needs (ECMEN) indicator which is sometimes used capture the economic vulnerability of households when aggregating CARI will not be tested in this study for a number of reasons. First, it requires the standard long expenditure module which cannot be collected over phone. Second, ECMEN calculation requires having an

updated value of the minimum expenditure basket (MEB) of the study population, which is not available for the areas where the study is being conducted.

It is important to clarify that the objective of this study is not to report national food security numbers t the national level or aim for external validity across the entire population of each pilot country. Instead, the primary goal is to assess the comparability of the two methodologies: i.e., the standard Consolidated Approach for Reporting Food Insecurity (CARI) with its remote version (rCARI) using a random sample drawn from the same population sample with comparable statistical characteristics. By focusing the study on the same population, it allows for a direct comparison of the effectiveness and accuracy of data collected remotely for reporting household food security.

The findings of this study is intended to provide valuable insights into how comparable F2F (CARI) and remote (rCARI) modalities (methodologies) are in reporting prevalence of food security. In this regards, conclusions drawn from this study should be interpreted within these caveats and context. Furthermore, while factors such as mobile phone ownership (mobile phone penetration rate), network coverage and survey response rate should be considered when interpreting remote surveys, because these may skew the study sample towards a particular group of respondents. The current study is not affected by these confounding factors for two reasons. First, the objective is not to report results at national level but to assess the effectiveness of remote modules and secondly, the sample used is uniform with comparable observable characteristics making them statistically. By emphasizing internal validity within the selected populations, this study aims to provide robust evidence to inform decision-making on the one side while improving food security assessments particularly in the context of remote data collection.

The rest of the report is structured as follows. In the next section, the study design and implementation is presented in detail. Section 3 presents the descriptives statistics of the data. Section 4 presents the results of comparative analysis showing how the different module results under the different survey modalities. Section 5 presents the regression results while 6 presents the summary and conclusions. Section outlines next steps including potentially some further analysis.

2 Study design & sample

The study was implemented as a survey experiment with stratified sample drawn from the same population with comparable statistics.

Selection of study areas and participating households followed a standard stratified random sampling approach (stratification was done by the different Admin Areas-biggest to smallest but not representative at each admin area) based on 3-point criteria - **mobile phone coverage, prevalence of food security & accessibility (considering the security situation on the ground)** - developed and agreed together with the WFP Burkina country office.

The study sample were then randomly selected randomly assigned to one of two groups: F2F or remote groups. The full process outlining the systematic steps (1 - 5) adopted for selecting the different administrative areas and respondent households with annotated explanations are shown in figure 1.

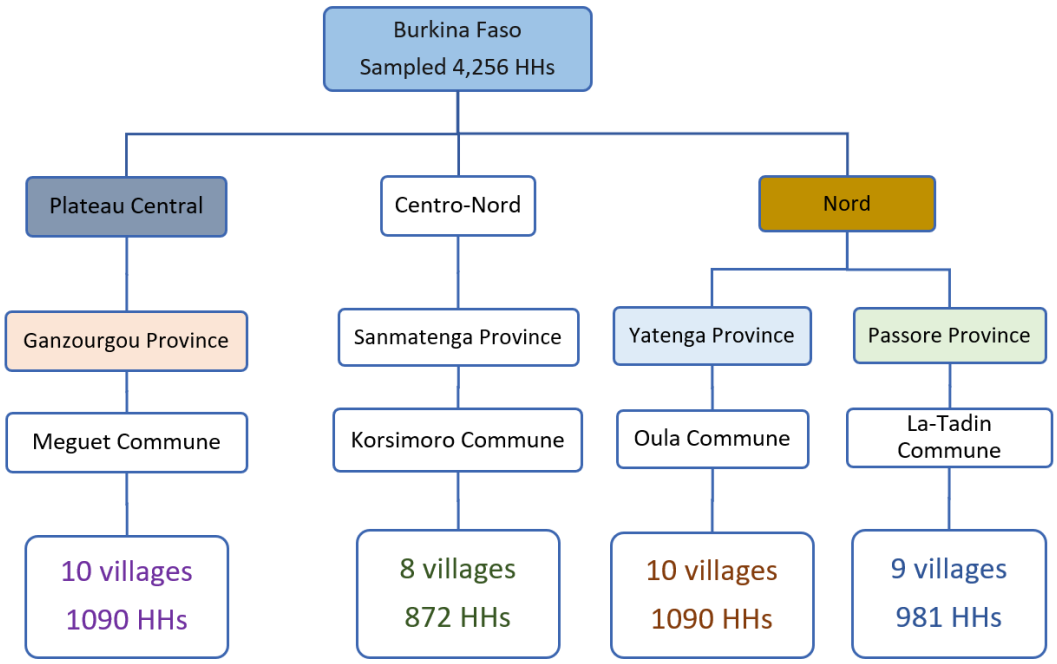

Figure 1: rCARI Validation Study Sampling Frame

1. STEP 1 - Based on the inclusion criteria (phone coverage, prevalence of food security and physical accessibility), three regions were purposively selected for the pilot in Burkina Faso (Plateau Central, Nord & Centro-Nord Regions).

2. STEP 2 - A total of four Provinces (ADMIN2), Ganzourgou, Sanmatenga, Yatenga and Passore, were purposively selected from the regions selected in step 1. The Administrative structure of Burkina Faso allowed for the inclusion of Admin3 areas (communes), therefore four communes were further selected; 2 communes from the Nord region and one each from

Centro-Nord and Plateau Central regions respectively.

3. STEP 3 - Selection of villages (ADMIN5): A list of villages in the selected communes (that met the inclusion criteria set out in STEP 1) were compiled at this stage to form the primary sampling units (PSUs). A total of 39 PSUs were initially selected (distributed uniformly across the 4 selected communes). The selection of the villages were done taking into consideration food security prevalence situation allowing for a mix of areas that would represent the different IPC phases in the country as a whole. See Annex Table 18 for the full list of the selected administrative areas selected, population and IPC phases.

4. STEP 4 - Sampling of households for the study: The sampling frame for study was all households in selected PSUs. A minimum of 100 households per PSU were randomly selected from the census list. Theses selected households were used in the randomization process in the next step.

5. STEP 5 - Assignment of household to survey modalities: Consented and sampled households from STEP 4 were randomly assigned into one of two survey modalities/groups; F2F (CARI group) and Remote (rCARI group). The experiment was designed such that the sample was randomly assigned equally between the two modalities.

   The interviews for both modalities - F2F and remote - were conducted within the same time frame to ensure households were reporting food security situation over the same window and also exposed to the same economic situations or shocks if any.

3 Descriptive statistics

In this section, the core descriptive statistics of the study sample are presented. Given the rather elaborate design that has been implemented, a number of statistics in the form of interview completion by modality, balance tables and attrition statistics across the different levels of implementation are presented.

3.1 Survey completion

Table 1 shows the breakdown of completed interviews by modality. A total of 2,882 interviews were completed out of 4,256, giving an overall completion rate of 67%. Of the 2,131 households sampled for face-to-face (F2F) interviews, 1,864 (87%) completed the survey, while of the 2,125 households sampled for remote interviews, 1,019 (47%) households completed the survey.

Table 1: Sample Completion: Face-to-Face vs. Remote

	Not Completed		Completed		Total	
	Count	Percent	Count	Percent	Count	Percent
Face to Face	268	12.58	1863	87.42	2131	100.00
Remote	1106	52.05	1019	47.95	2125	100.00
Total	1374	32.28	2882	67.72	4256	100.00

The rainy season, during which most respondents - primarily farmers - were busy working in the fields, along with poor phone network coverage in many study areas, impacted respondents' availability for interviews.

3.2 Modality balance check for the completed sample

A key requirement of the study design is the statistical comparability of the two study groups at the time of data collection, assessed through a balance test based on observable characteristics. To evaluate the comparability (or equivalence) of the F2F and Remote study groups, baseline demographics and other observable characteristics collected during the household census phase were analyzed using appropriate statistical tools (test of group means - student t-test, chi square). This comparison of means between the two modalities allowed for the assessment of statistical equivalence of the two groups (F2F and remote) based on the number of households that have completed the interviews.

Additional statistical balance tests, based on data collected during the census phase, were conducted and are presented in the Annex. These tests assessed balance across different levels of randomization—specifically, modality and form length—using the originally sampled households that were randomly assigned to the various groups.

Table 2 presents the results of the baseline variable comparison across the two modalities. Except for the variable indicating whether the household head is the respondent, the households are generally comparable across the two modalities based on observable baseline characteristics.

Table 2: Balance Table by Randomized Survey Modality: Face-to-Face vs. Remote for Completed Surveys

Variable	F2F	Total Remote	Diff
List respondent age	45.48	45.65	0.16
	(16.79)	(16.31)	(0.80)
Household head is the respondent	0.77	0.82	0.05**
	(0.42)	(0.39)	(0.00)
Female Household Head	0.20	0.19	-0.01
	(0.40)	(0.39)	(0.41)
Married Household Head	0.98	0.99	0.01
	(0.15)	(0.12)	(0.07)
Household head years of education	1.42	1.42	-0.00
	(2.92)	(2.94)	(1.00)
Displaced household	0.03	0.04	0.01
	(0.17)	(0.20)	(0.13)
Household size	7.11	7.42	0.31
	(4.36)	(4.36)	(0.07)
Number of household member younger than 15	3.14	3.26	0.12
	(2.42)	(2.39)	(0.19)
Number of household member older than 60	0.58	0.64	0.06
	(1.07)	(1.04)	(0.18)
Has household member with disability/chronic disease	0.18	0.17	-0.01
	(0.38)	(0.37)	(0.36)
Number of rooms	1.69	1.68	-0.01
	(1.17)	(1.20)	(0.81)
Household Crowding Index	4.92	5.21	0.30
	(3.77)	(4.02)	(0.05)
Household has impoverished roof	0.02	0.01	-0.00
	(0.12)	(0.12)	(0.95)
Household has impoverished floor	0.15	0.17	0.02
	(0.36)	(0.38)	(0.18)
Observations	1,863	1,019	2,882

Note: Robust standard deviations in parentheses; * * p<0.05, ** p<0.01, *** p<0.001.

The statistical comparison of the two modalities, based on baseline observable characteristics collected during the household census, indicates that the groups are statistically balanced or equivalent. This provides a strong foundation to proceed with direct comparisons of outcome indicators across the two modalities to assess the effect of modality on these indicators. The results of these comparisons of the key indicators are presented in the following section. First, the main rCARI and CARI aggregations are methods are tested based on the modality that the indicators were calculated. Second, the individual indicators - food consumption score (FCS), reduced coping strategy index (rCSI), livelihood coping strategy index (LCSI), food expenditure score (FES) and income source and income change - are tested over modalities. Post-survey attrition analysis can be found in the attrition tables in the Annex using the attrition data for each modality.

4 Results & discussions

In this section, the main results of the study aggregated by the food security indicator are analysed and presented. Here, the different versions of CARI calculated using F2F or remote indicators are compared with the standard CARI. A test of group means using student t-test of equivalence is performed to determine the effect of modality on any difference between any two methodologies that are tested at a time.

4.1 CARI vs. rCARI

Conventional/standard CARI: CARI-FES: Standard CARI using FES from F2F surveys.

The other methods of calculating and aggregating CARI (that have been tested)

1. **rCARI-FES:** CARI using FES from remote surveys.
2. **CARI-Inc:** CARI using the income variable from F2F surveys.
3. **rCARI-Inc:** rCARI using income source and income change variable collected from remote surveys (this is the version of CARI that was calculated during the Covid-19 pandemic).

Table 3 presents the different combination of indicators used in each CARI aggregation methodology.

Table 3: Matrix of the different aggregation methods and the indicators used for each methodology

	Aggregation Methodology			
Indicator Included	**CARI (F2F)**	**CARI (F2F)**	**rCARI (Remote)**	**rCARI (Remote)**
FCS	Yes	Yes	Yes	Yes
rCSI	Yes	Yes	Yes	Yes
LCSI	Yes	Yes	Yes	Yes
FES	Yes	**No**	Yes	**No**
Income source & income change	**No**	Yes	**No**	Yes

The first comparison is between the results of the conventional CARI methodology (i.e., CARI using F2F modules) with the results of the CARI methodology aggregated using remote modules and replacing FES with income source and income change variable (i.e., rCARI-Inc). The results of this comparison are presented in Table 4.

The results show that, rCARI-Inc consistently over classified respondents as food secured - based on the CARI categories of food secure, marginally food secure and moderately food insecure - by between 4% and 13% points while underreporting the prevalence of severely food insecure category by up to 27% points. In terms of the overall Acute Food Insecurity (AFI) (i.e., combination of the moderately food insecure and severely food insecure categories), the remote modality using the income source and income change indicator tends to under reports the severity by up to 15 percentage points compared to the standard F2F CARI (AFI being 89% for CARI and 74% for

rCARI respectively).

Table 4: Impact of Survey Modality: Standard CARI-FES vs. Standard rCARI-Inc

	Burkina Faso		
Variable	F2F	Remote	Diff
Food Secure	0.01	0.05	0.04***
	(0.11)	(0.23)	(0.00)
Marginally Food Secure	0.10	0.20	0.10***
	(0.30)	(0.40)	(0.00)
Moderately Food Insecure	0.59	0.72	0.13***
	(0.49)	(0.45)	(0.00)
Severely Food Insecure	0.29	0.03	-0.27***
	(0.46)	(0.16)	(0.00)
Food Insecure Group	0.89	0.74	-0.15***
	(0.32)	(0.44)	(0.00)
Observations	1,863	1,019	2,882

Note: Robust standard deviations in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

From the results presented in Table 4, it can be concluded that rCARI-Inc methodology tends to under-report severe food insecurity while overestimating food security prevalences. This suggests that households at the lower end of the food security spectrum may be under represented when using the income module in phone-based food security assessments compared to the expenditure module in F2F assessments.

In the next combination the different consoles are compared. Table 5 compares the results CARI using a shorter expenditure version which was implemented over the phone. This module unlike the standard expenditure module only collects the household's aggregated expenditure items covering both food and non-food expenditure and has 23 questions (An extensive explanation of this module (short & long modules) will be covered in later sections of this report).

The results of this version (rCARI-FES) show no significant statistical difference when compared with the results of the standard CARI F2F methodology. However, the absolute percentages show that rCARI-FES consistently underestimates the prevalence of food insecurity across all the 4 CARI categories. Thus, it can be concluded that the shorter expenditure module when used in remote assessments produces results that are statistically comparable to the standard F2F CARI that uses the full expenditure module. The implication of this result is that, the shortened expenditure module (rCARI-FES) provides a feasible and relatively comparable alternative to the standard F2F expenditure module to assess economic vulnerability in remote surveys. While it may slightly underestimates food insecurity prevalence, the lack of statistical significance suggests that it can serve as a viable substitute when F2F assessments are not practical. However, recognizing and accounting for its conservative numbers is critical to ensure the results are reliable and actionable.

Being a new indicator that is being tested, the income source and income change module was also collected in the F2F modality. This allowed us to compare the results of the same module over the two modalities. These results are presented in Table 6 comparing the CARI F2F and rCARI both using the income source and income indicator collected in F2F and over the phone

Table 5: Impact of Survey Modality: CARI-FES vs. rCARI-FES

	Burkina Faso		
Variable	F2F	Remote	Diff
Food Secure	0.01	0.03	0.01*
	(0.11)	(0.16)	(0.01)
Marginally Food Secure	0.10	0.09	-0.01
	(0.30)	(0.29)	(0.37)
Moderately Food Insecure	0.59	0.62	0.02
	(0.49)	(0.49)	(0.21)
Severely Food Insecure	0.29	0.27	-0.03
	(0.46)	(0.44)	(0.11)
Food Insecure Group	0.89	0.88	-0.00
	(0.32)	(0.32)	(0.75)
Observations	1,863	1,019	2,882

Note: Robust standard deviations in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

respectively. From the results, two key findings are deduced. On the one hand, rCARI (remote) classifies significantly fewer households as severely food insecure compared to the CARI (F2F) - (7% less) - while on the other hand, in terms of AFI prevalence, the results show no statistical significance between the numbers produced by the two methodologies.

This suggests that while using the income indicator in remote surveys may underestimate the severity of food insecurity at the extreme end, it's ability to generally classify the overall AFI, i.e., identifying households classified as moderately or severely food insecure is comparable across survey modalities. This insight is important for understanding the limitations and comparability of using the income source and income choice indicator in food security assessments particularly remote assessments.

Table 6: Impact of Survey Modality: CARI-Inc vs. rCARI-Inc

	Burkina Faso		
Variable	F2F	Remote	Diff
Food Secure	0.05	0.05	0.00
	(0.22)	(0.23)	(0.63)
Marginally Food Secure	0.18	0.20	0.02
	(0.38)	(0.40)	(0.22)
Moderately Food Insecure	0.68	0.72	0.04*
	(0.47)	(0.45)	(0.02)
Severely Food Insecure	0.09	0.03	-0.07***
	(0.29)	(0.16)	(0.00)
Food Insecure Group	0.77	0.74	-0.03
	(0.42)	(0.44)	(0.06)
Observations	1,863	1,019	2,882

Note: Robust standard deviations in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

The next analysis assesses how households (same resondent) reports their food security based on two economic vulnerability indicators, i.e., expenditure (FES) and income source and change. In this analysis, a paired sample t-test was used to evaluate the results of the results. This was done separately for the F2F and remote samples respectively. Table 7 presents the results for the F2F

sample while Table 8 shows the results for the remote sample.

Table 7 shows the results for the paired responses in the F2F sample. More households tend to report better food security outcomes with the income indicator (higher number of households being classified as food secure, marginally food secure, moderately food insecure, between 4 and 9 percentage points) than the FES. However, the reverse is found to be true for the severely food insecure category where more households are captured under the FES than is captured by the income source and income change indicator. Aggregating to the overall AFI prevalence show that the income indicator is likely to under estimate AFI numbers by 17 percentage points compared to the FES indicator. This suggests that households considered as acute food insecure under the FES indicator would likely be reclassified as food secure when using the income-based indicator in its current form. These differences highlight how the two economic vulnerability indicators - FES & income indicator are likely to impact food security numbers due to the current categorization of the income groupings.

Table 7: CARI: FES vs. Inc - Paired Responses (F2F Surveys)

	Burkina Faso		
Variable	F2F	Remote	Diff
Food Secure	0.01	0.05	-0.04***
	(0.01)	(0.01)	(0.01)
Marginally Food Secure	0.10	0.18	-0.08***
	(0.01)	(0.01)	(0.01)
Moderately Food Insecure	0.59	0.68	-0.09***
	(0.02)	(0.02)	(0.02)
Severely Food Insecure	0.31	0.09	0.21***
	(0.01)	(0.01)	(0.01)
Food Insecure Group	0.89	0.78	0.17***
	(0.01)	(0.01)	(0.01)
Observations	1,737	1,737	1,737

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Table 8 presents the results of the paired response from the remote sample (same households and respondents). The results are strikingly similar to what has been observed with the F2F sample reported in Table 7.

Thus, the overall conclusion drawn from the two tables (Tables 7 & 8) is that regardless of the modality of the survey administration, the income source and income change indicator tended to significantly underestimate the prevalence of AFI by 14 and 17 percentage points on aggregate.

4.2 Sensitivity analysis of the Income source & income change vs. FES classifications

The results comparing different versions of rCARI to the standard F2F CARI so far indicates that, any significant difference that is observed is driven by the choice of economic vulnerability indicator used, i.e., either the FES or income source and income change indicators. Therefore, in this section, a further sensitivity analysis is conducted to see how the CARI categories using the

Table 8: rCARI: FES vs. Income - Paired Responses (Remote Surveys)

	Burkina Faso		
Variable	F2F	Remote	Diff
Food Secure	0.03	0.05	-0.03***
	(0.01)	(0.01)	(0.01)
Marginally Food Secure	0.09	0.20	-0.12***
	(0.01)	(0.01)	(0.01)
Moderately Food Insecure	0.62	0.72	-0.10***
	(0.02)	(0.02)	(0.02)
Severely Food Insecure	0.27	0.03	0.24***
	(0.02)	(0.02)	(0.02)
Food Insecure Group	0.88	0.74	0.14***
	(0.01)	(0.01)	(0.01)
Observations	1,007	1,007	1,007

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

income indicator compares with CARI FES categories in terms of classifying food security. The cross-tabulation of the four-category CARI classifications using income and FES provides nuanced insights into how households are categorized across different food security tiers.

4.2.1 Sensitivity of the 4-CARI categories based on Income & FES

F2F Sample

Table 9: Cross Tabulation of CARI_Income and CARI_FES - F2F Sample

	FES									
	Food Secure		Marginally Food Secure		Moderately Food Insecure		Severely Food Insecure		Total	
	Count	Percent	Count	Percent	Count	Percent	Count	Percent	Count	Percent
Food Secure	23	25.00	69	75.00	0	0.00	0	0.00	92	100.00
Marginally Food Secure	0	0.00	114	34.23	219	65.77	0	0.00	333	100.00
Moderately Food Insecure	0	0.00	6	0.48	854	67.62	403	31.91	1263	100.00
Severely Food Insecure	0	0.00	0	0.00	29	16.57	146	83.43	175	100.00
Total	23	1.23	189	10.14	1102	59.15	549	29.47	1863	100.00

Table 9 presents a cross-tabulation of food security classifications derived from income and FES CARI categories from the F2F sample. The results highlight significant alignment in the classification severely food insecure households. For example, Table 9 show that of the households classified as severely food insecure by the income source and income change indicator (n = 175), 83.43% of these households were also captured and classified as severely food insecure using the FES indicator with the remaining 16.57% of households being classified as moderately food insecure by the FES. In terms of the moderately food insecure classification, of the households that were classified by the income choice and income change indicator (n = 1,236), 68% were also classified as moderately food insecure while 32% were classified as severely food insecure by the FES indicator.

The results for the higher food security categories on the other hand show significance divergence between the two aggregating methodologies. For example, up to 7 in 10 of all the households are classified as food secure or marginally food secure are classified into a less food secure category by FES with the alignment being found in about 30% of all classifications in these categories.

- Severe food insecurity is consistently captured by both measures, indicating robust alignment

at the lower end of the food security spectrum.

- CARI using income source and income change indicator may overestimate food security particularly amongst populations with borderline food security, risking exclusion errors in assistance targeting.

The overarching conclusion of Table 9 shows a strong consistency between income choice and income change and FES based CARI classifications, particularly for severely and moderately food insecure households. This alignment though not perfect provides evidence to support the use of the income based CARI (rCARI) as an effectiveness methodology for identifying vulnerable and food insecure populations. However, in less severe or transitional populations, the two methodologies diverge up to 75%.

Remote Sample

Table 10: Cross Tabulation of CARI_Income and CARI_FES - Remote Sample

	FES									
	Food Secure		Marginally Food Secure		Moderately Food Insecure		Severely Food Insecure		Total	
	Count	Percent	Count	Percent	Count	Percent	Count	Percent	Count	Percent
Food Secure	26	48.15	28	51.85	0	0.00	0	0.00	54	100.00
Marginally Food Secure	1	0.50	61	30.65	137	68.84	0	0.00	199	100.00
Moderately Food Insecure	0	0.00	2	0.28	481	66.25	243	33.47	726	100.00
Severely Food Insecure	0	0.00	0	0.00	4	14.29	24	85.71	28	100.00
Total	27	2.68	91	9.04	622	61.77	267	26.51	1007	100.00

Table 10 presents the results for the sensitivity analysis conducted on the remote sample. The results follow similar trend of what has been reported in Table 9 further reinforcing the fact that the income source and income change indicator largely masks vulnerabilities especially along the food secure and borderline populations, while showing large convergence (up 85% agreement between income and FES) for the severely food insecure groups.

4.2.2 Sensitivity of food secure vs. food insecure grouping to income & FES

Table 11 presents the results as is used for targeting purposes where moderately and severely food insecure categories are considered together as acute food insecure and thus fall into the assisted category. In terms of the food insecure group, both indicators show a strong consistency. For example, of the total sampled households, 1,651 (88.6%) were classified as AFI (vulnerable) using FES. Out of this number, 1,432(86.7%) of these households were also simultaneously captured as experiencing AFI with 219(13.3%) were misclassified as food secure using the income source and income change indicator. However, among those income-based food secure (425), a little over half of them 219 (51.5%) were classified as being AFI by FES, meaning nearly half of those not vulnerable by income source and income indicator were flagged as being vulnerable based on how much of their expenditure go to food needs.

Generally, these results show a strong concordance between the two indicators in terms of identifying vulnerability however the FES seems to be more sensitive to capturing broader dimensions which could be missed by the income source indicator. The FES may be able to flag hidden food insecurity

among households considered not be engaging in low income activities but dedicating large shares of expenditure to food thus potentially indicating trade-offs affecting their overall food security situation.

Table 11: Cross Tabulation of Acute Food Insecurity (CAR_Income vs. CARI_FES) - F2F Sample

	Food insecure with FES		Food secure with FES		Total	
	Count	Percent	Count	Percent	Count	Percent
Food insecure with Income	1432	99.58	6	0.42	1438	100.00
Food secure with Income	219	51.53	206	48.47	425	100.00
Total	1651	88.62	212	11.38	1863	100.00

Similar to Table 11, the results in Table 12 follow the same trend using the remote sample. Of the households classified as AFI using the FES (899 households), 762(84.8%) of these households were also captured as facing AFI using the income source indicator. While for the households classified as by the income AFI, almmost all (99.5%) households were classified as AFI by the FES indicator indicating a strong overlap and consistent with what has been observed in the F2F sample (Table 11). Interestingly, among the households (n = 253) classified as being food secure using the income source indicator 54.15% of them were captured as being food insecure using the FES indicator, again revealing significant misalignment for the upper class households.

Table 12: Cross Tabulation of Acute Food Insecurity (CAR_Income vs. CARI_FES) Remote Sample

	Food insecure with FES		Food secure with FES		Total	
	Count	Percent	Count	Percent	Count	Percent
Food insecure with Income	762	99.48	4	0.52	766	100.00
Food secure with Income	137	54.15	116	45.85	253	100.00
Total	899	88.22	120	11.78	1019	100.00

The conclusion though the results of Tables 11 and 12 show a a high level of consistency in identifying food-insecure households, the reverse is true for the non poor households where income source may wrongly classify households as food secure although they allocate a lot of resources to purchasing food. The income-based food security classifications may tend to mask food insecurity by misclassifying up to 50% of food-insecure households (particularly if their income source is considered stable and not changing) as being food secure. Further analysis of the data to understand this trend and realign the income indicator to better reflect these nuances of these income sources to better capture and reflect the results of the FES indicator is important.

4.3 Comparison of the food security indicators across modality

In this section, the overarching objective is to evaluate each individual food security indicator statistically across the two data collection modalities (face-to-face and remote) to identify similarities or significant differences. By doing so, the analysis aims to determine which indicators yield con-

sistent results across both modalities, indicating strong comparability, and which indicators show substantial variation, suggesting that the data collection method (F2F vs. remote) may influence reported outcomes.

Understanding these comparisons helps identify which indicators are robust to modality effects, supporting their reliability for use in remote data collection contexts, and which indicators may require adjustment or careful interpretation when collected remotely.

4.3.1 Food Consumption Score (FCS)

The Food Consumption Score (FCS) is a composite score based on households' dietary diversity, food consumption frequency, and relative nutritional importance of different food groups, and is calculated by inspecting how often households consume food items from the different food groups during a 7-day reference period. The responses are aggregated, and weights applied to obtain an index which is then used to categorize the household into one of three groups of food consumption: food secure, borderline, and poor.

Table 13: Impact of Survey Modality on Food Consumption Score and Food Groups

	Burkina Faso		
Variable	F2F	Remote	Diff
Food Consumption Score	26.26	27.15	0.89*
	(10.65)	(9.08)	(0.02)
FCS: Acceptable	0.08	0.07	-0.02
	(0.27)	(0.25)	(0.13)
FCS: Borderline	0.24	0.23	-0.01
	(0.43)	(0.42)	(0.68)
FCS: Poor	0.68	0.70	0.02
	(0.47)	(0.46)	(0.22)
Consumption over the past 7 days (cereals and tubers)	6.54	6.93	0.39***
	(1.13)	(0.37)	(0.00)
Consumption over the past 7 days (pulses)	1.06	1.00	-0.06
	(1.31)	(1.38)	(0.25)
Consumption over the past 7 days (dairy products)	0.31	0.26	-0.05
	(1.13)	(1.11)	(0.25)
Consumption over the past 7 days (protein-rich foods)	0.57	0.26	-0.30***
	(1.16)	(0.86)	(0.00)
Consumption over the past 7 days (vegetables)	3.23	5.56	2.32***
	(2.98)	(1.57)	(0.00)
Consumption over the past 7 days (fruit)	0.09	0.08	-0.02
	(0.59)	(0.52)	(0.46)
Consumption over the past 7 days (oil)	2.48	1.88	-0.60***
	(2.02)	(1.87)	(0.00)
Consumption over the past 7 days (sugar)	3.89	3.28	-0.61***
	(2.72)	(2.80)	(0.00)
Observations	1,863	1,019	2,882

Note: Robust standard deviations in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Table 11 show results of the comparison of the FCS collected and reported over the two modalities. The over all picture drawn from the results presented show that there is no marked difference between the aggregated FCS reported by respondents over the phone and in the F2F surveys respectively. However, a deeper analysis of the individual food groups show that respondents reported consuming significantly more vegetables, cereals and tubers than F2F households. On the

other hand, F2F households were more likely to report higher consumption of protein rich foods, oil and sugar compared to the remote sample.

4.3.2 Reduced Coping Strategies Index (rCSI)

The reduced Coping Strategies index (rCSI) is used to assess the frequency of occurrence of increasingly severe coping strategies within a households' due to lack of adequate food, i.e., the behaviours individuals engage in when they can not access enough food. It measures the frequency and severity of the food consumption related behaviours by households to cope with food shortage within a 7-day recall period. The indicator is comprised of 5 standard strategies framed around how and who consumes food in times of shortage. It serves as a proxy indicator for food access. The higher the composite score, the more coping reported, and therefore the more food insecure the household is.

Table 12 presents the results of the rCSI module compared over the two samples and modalities. The results show that consistently respondents over the phone reported significantly lower rCSI compared to those who responded in the F2F interviews. This means that households tend to report extensive use of negative coping strategies in the F2F survey than they do on the phone. The difference in the overall rCSI between the two modalities is about 3 points and is statistically significant.

Table 14: Impact of Survey Modality on Reduced Coping Strategy Index Classification

	Burkina Faso		
Variable	F2F	Remote	Diff
Reduced Coping Strategies Index (rCSI)	9.72	6.86	-2.86***
	(10.23)	(7.89)	(0.00)
rCSI: Low	0.38	0.48	0.10***
	(0.49)	(0.50)	(0.00)
rCSI: Median	0.43	0.42	-0.00
	(0.49)	(0.49)	(0.84)
rCSI: Severe	0.19	0.10	-0.10***
	(0.40)	(0.29)	(0.00)
Relied on less preferred, less expensive food	1.85	1.02	-0.83***
	(2.16)	(1.53)	(0.00)
Borrowed food or relied on help from friends or relatives	0.50	0.37	-0.13***
	(1.08)	(0.88)	(0.00)
Reduced portion size of meals at meals time	1.93	1.13	-0.80***
	(2.40)	(1.78)	(0.00)
Restricted consumption by adults in order for young-children to eat	1.14	1.14	-0.00
	(1.71)	(1.58)	(1.00)
Reduced the number of meals eaten per day	1.52	0.55	-0.97***
	(2.14)	(1.30)	(0.00)
Observations	1,863	1,019	2,882

Note: Robust standard deviations in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

4.3.3 Livelihood Coping Strategies Index (LCSI)

The Livelihood Coping Strategy (LCSI) indicator measures a household's ability to cope with food insecurity due to food shortages or insufficient funds over a medium- to long-term period (30 days to 12 months). It assesses the household's strategies to meet dietary needs when faced with economic or food-related challenges. LCSI strategies are typically categorized into three levels: stress, crisis, and emergency. For targeting and assistance, only the crisis and emergency categories are generally

considered, as they classify households as food insecure.

Table 13 presents the aggregated LCSI results across the four main categories. The data indicates that households in the remote survey report better coping compared to those in the F2F module. For example, nearly half (50%) of households in the remote survey reported not adopting any coping strategies, compared to 43% in the F2F survey. In terms of crisis and emergency strategies, 28% of F2F respondents adopted these strategies, compared to 19% of remote respondents.

While the remote sample shows a higher proportion of households not having adopted any coping strategies, the differences in the adoption of crisis and emergency strategies between the two groups (28% for F2F vs. 19% for remote) suggest a trend that warrants further investigation. The remote sample tend to significantly adopt more of crisis strategies (4% points more) and less of emergency level strategies (13% points less) overall compared to the F2F sample.

Table 15: Impact of Survey Modality on Livelihood Coping Strategy

	Burkina Faso		
Variable	F2F	Remote	Diff
Maximum Livelihood Coping: None	0.43	0.49	0.06***
	(0.49)	(0.50)	(0.00)
Maximum Livelihood Coping: Stress level	0.29	0.31	0.02
	(0.45)	(0.46)	(0.29)
Maximum Livelihood Coping: Crisis level	0.03	0.07	0.04***
	(0.17)	(0.26)	(0.00)
Maximum Livelihood Coping: Emergency level	0.25	0.12	-0.13***
	(0.43)	(0.33)	(0.00)
Observations	1,863	1,019	2,882

Note: Robust standard deviations in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

To better understand the composition of the categories reported in Table 13, the individual coping strategies are disaggregated and presented in Table 14. This breakdown allows for a more detailed analysis of the specific strategies households are adopting. The results in Table 14 confirm the findings from Table 13, where households in the remote sample consistently report adopting more extreme coping strategies, such as crisis or emergency measures, compared to the F2F sample suggesting a higher levels of vulnerability amongst the remote sample.

4.3.4 Food Expenditure Share (FES)

The Food Expenditure Share (FES) is a key indicator of food security, reflecting the proportion of a household's total consumption expenditure spent on food. A higher FES suggests greater economic vulnerability, as a larger portion of household resources is dedicated to food, potentially limiting spending on other necessities.

Table 15 summarizes the FES across two survey modalities. This includes results from the shortened expenditure module conducted remotely, which was designed as a pilot to assess the feasibility of collecting expenditure remotely. The comparison between the remote and F2F surveys reveals no

Table 16: Distribution of Individual Coping Strategies Adopted

	Burkina Faso		
Variable	F2F	Remote	Diff
Used Stress coping: Sold household assets/goods	0.05	0.02	-0.02***
	(0.21)	(0.15)	(0.00)
Used Stress coping: Reduced expenses on health and education	0.17	0.14	-0.03*
	(0.38)	(0.35)	(0.02)
Used Stress coping: Spent savings	0.30	0.22	-0.08***
	(0.46)	(0.41)	(0.00)
Used Stress coping: Borrowed money	0.29	0.27	-0.02
	(0.45)	(0.45)	(0.27)
Used Crisis coping: Sold productive assets or means of transport	0.02	0.04	0.02**
	(0.14)	(0.19)	(0.00)
Used Crisis coping: Moved to somewhere else	0.01	0.06	0.05***
	(0.11)	(0.24)	(0.00)
Used Crisis coping: Children worked	0.04	0.06	0.02
	(0.20)	(0.23)	(0.07)
Used Emergency coping: Sold houses	0.00	0.01	0.01*
	(0.05)	(0.10)	(0.01)
Used Emergency coping: Begged and/or scavenged	0.05	0.03	-0.02**
	(0.22)	(0.17)	(0.00)
Used Emergency coping: Sold female animal	0.21	0.10	-0.11***
	(0.41)	(0.30)	(0.00)
Observations	1,863	1,019	2,882

Note: Robust standard deviations in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

significant differences in the aggregated FES between the two methods.

The FES is typically grouped into four standard categories for analysis: less than 50%, 50-75%, 75% or more, and a non-food expenditure category. When comparing these categories across survey methods, mixed results emerge. Notably, a significantly larger proportion of F2F respondents report food expenditure shares between 50% and 75% of total household spending. In contrast, the remote survey sample shows a different distribution, with a majority of respondents reporting either very low (less than 50%) or very high (75% or more) FES.

Statistical tests comparing the results of the two outcomes (short FES-remote and the standard FES-F2F) show that, the aggregated FES values from the remote surveys do not differ significantly from the results if the F2F, suggesting that a potential for the shorter FES indicator to be streamlined and implemented over the phone to collect economic vulnerability indicator for food security assessment. However, it must be noted that when breaking down the FES into the four categories, the distribution of responses varies considerably between the two modalities. For example, the data show that when using the short version of the FES module (remote modality), 53% of households report food expenditure shares of 75% or more while 19% of households reported having a FES of less than 50%. This is significantly different from the 41% and 14% of households for the same FES categories respectively when using the standard expenditure module in the F2F.

The expenditure analysis presented in Table 15 reveals significant differences in household classifications across the four CARI food expenditure share groups, which assess the severity of food insecurity based on shares of expenditure spent on food. However, the aggregated FES data show no significant difference between the shortened remote expenditure module and the standard ex-

Table 17: Impact of Survey Modality on Food Expenditure Share

	Burkina Faso		
Variable	F2F	Remote	Diff
Food Expenditure Share	0.69	0.67	-0.01
	(0.17)	(0.28)	(0.12)
FES < 50 percent	0.14	0.19	0.05***
	(0.34)	(0.39)	(0.00)
FES: 50-65 percent	0.24	0.13	-0.11***
	(0.43)	(0.34)	(0.00)
FES: 65-75 percent	0.21	0.15	-0.06***
	(0.41)	(0.36)	(0.00)
FES: >= 75 percent	0.41	0.53	0.12***
	(0.49)	(0.50)	(0.00)
Observations	1,863	1,019	2,882

Note: Robust standard deviations in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

penditure module in F2F. This is largely due to the distributional effect of the group differences that tend to cancel out in the aggregation.

4.3.5 Income Source and Income Change

The income indicator captures the respondent's or household's income sources and changes in income over the past 12 months. Introduced during the COVID-19 pandemic to support remote assessments, it serves as a proxy for economic vulnerability when FES data cannot be collected. This was particularly important during the pandemic when expenditure data was difficult to gather. The indicator considers the impact of economic changes—both positive and negative—on household income, providing a measure of vulnerability. The results are presented in Table 16.

The results of the income source and income change indicator, as compared across modalities, show that respondents tend to under-report their food security in remote surveys. Specifically, a higher proportion of households reported having a stable, regular income source in the F2F interviews compared to the remote surveys.

Table 18: Impact of Survey Modality on Income

	Burkina Faso		
Variable	F2F	Remote	Diff
Income: Regular and Unchanged/Increasing	0.58	0.82	0.23***
	(0.49)	(0.39)	(0.00)
Income: Regular and Reduced OR Irregular and Unchanged/Increasing	0.37	0.17	-0.20***
	(0.48)	(0.38)	(0.00)
Income: Irregular and Reduced	0.02	0.00	-0.02***
	(0.15)	(0.04)	(0.00)
Income: No income sources	0.02	0.01	-0.02***
	(0.15)	(0.08)	(0.00)
Observations	1,863	1,019	2,882

Note: Robust standard deviations in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

5 Regression analysis

In this section of the analysis, a simple linear regression analysis is conducted to determine the effect of survey modality and phone ownership in explaining the food security (defined as moderately and severely food insecure based on CARI) of the household controlling for individual and household level characteristics. The model estimated is specified as follows:

$$Y_i = \beta M_i + \gamma P_i + \delta X_i + \varepsilon_i \tag{1}$$

where Y_i is a binary indicator of the food security status of the ith household taking the value of one if the respondent is food insecure (made of the moderately food insecure plus severely food insecure CARI categories) and zero otherwise (calculated independently for CARI-FES - Food Insecure FES, and CARI-Inc - Food Insecure Income, indicators respectively), X_i is a vector of respondent's and household level characteristics such as age of respondent, sex, education in years, household size, dependency ratio and presence of persons with disability in the household, M_i is a binary indicator $M_i = 0/1$ taking the value of 1 if the mode of interview was F2F and zero if phone, P_i is phone ownership indicator of the respondent $P_i = 1/0$ takes the value of one if the respondent owned a phone zero otherwise and ε_i is the error term clustered at ADMIN5 level. β and γ are the coefficients of interest showing the effect of modality and phone ownership on the ith household's food insecurity outcome respectively.

Equation (1) is implemented as a simple logistic regression for the full sample stepwise in three different configuration of the model defined by the number covariates included in each configuration. The results of the analysis are presented in Table 17 below. In the first stage, a base model which includes only the survey modality and gender of the respondent (columns 1 & 4). The second model includes the base model controlling for additional baseline household demographics collected in the census exercise (columns 2 & 5). The third and final estimation is the second model plus an interaction term of the sex of the household head and sex of the respondent (columns 3 & 6).

The key variable of interest is the survey modality (remote=1). The coefficient for the survey modality (Columns 1-3) is 0.00, with a standard error of (0.02), indicating no significant relationship between survey modality and the likelihood of being classified as food insecure when using the FES indicator in the F2F surveys suggesting that CARI classification based on the use of FES is not strongly affected by whether the data was collected through F2F or remotely by phone. The coefficient for modality (Columns 4-6) is 0.02, with a standard error of (0.03) for the CARI using income source and income change indicator, again indicating no significant relationship. The lack of significant differences suggests that both indicators (FES and Income) are relatively robust across survey modalities. However, FES shows slightly less variability compared to income, indicating it may be a more stable indicator across different survey methods.

The gender of the respondent a key indicator of food insecurity is found to be significantly associated

Table 19: Logit Regression on Food Security based Outcomes

	Food Insecure with FES			Food Insecure with Income		
	(1)	(2)	(3)	(4)	(5)	(6)
Survey modality (Remote=1)	-0.00	0.00	-0.00	-0.03	-0.02	-0.02
	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)
Sex of respondent		-0.05**	-0.05**		-0.08**	-0.09***
		(0.02)	(0.02)		(0.03)	(0.03)
List respondent age		-0.00	-0.00		-0.00	-0.00
		(0.00)	(0.00)		(0.00)	(0.00)
Household head is the respondent		-0.01	-0.01		0.01	0.01
		(0.01)	(0.01)		(0.02)	(0.02)
Female Household Head		0.02	0.02		0.03	0.01
		(0.02)	(0.03)		(0.02)	(0.04)
Married Household Head		0.02	0.02		-0.05	-0.05
		(0.03)	(0.03)		(0.06)	(0.06)
Household head years of education		0.00	0.00		0.00	0.00
		(0.00)	(0.00)		(0.00)	(0.00)
Household size		0.00	0.00		0.00	0.00
		(0.00)	(0.00)		(0.00)	(0.00)
Number of household member younger than 15		-0.01	-0.01		-0.00	-0.00
		(0.00)	(0.00)		(0.01)	(0.01)
Number of household member older than 60		0.01	0.01		-0.00	-0.00
		(0.01)	(0.01)		(0.01)	(0.01)
Has household member with disability/chronic disease		-0.02	-0.02		0.01	0.01
		(0.01)	(0.01)		(0.02)	(0.02)
Household Crowding Index		-0.00	-0.00		0.00	0.00
		(0.00)	(0.00)		(0.00)	(0.00)
Household has impoverished roof		-0.05	-0.05		-0.12	-0.12
		(0.05)	(0.05)		(0.10)	(0.10)
Household has impoverished floor		0.01	0.01		0.00	0.00
		(0.01)	(0.01)		(0.03)	(0.03)
RespSex_HHSex			-0.00			0.04
			(0.03)			(0.04)
Obs.	2882	2882	2882	2882	2882	2882

Note: Estimates are marginal effects using logit model regression. Robust standard errors are clustered at village level (Admin5) and reported in parentheses. * denotes statistical significance at 0.05 ** at 0.01 and *** at 0.001.

with AFI outcomes. For example, female respondents are between 5–6 percentage points less likely to be classified as food insecure compared to male respondents, and this relationship is statistically significant ($p < 0.05$) when using the FES indicator while they are up to 8 percentage points less likely to report food insecurity compared to male respondents using the income source indicator, with a higher statistical significance ($p < 0.01$).

This results suggest that female respondents were significantly less likely to be classified as food secure consistently across both CARI-Inc and CARI-FES methodologies compared to males. The inclusion of an interaction term between sex of the respondent and sex of the household did not show any statistical significance. This may reflect:

- Differences in how males and female respondents perceive or report income changes and food expenditure.
- Gender-specific income vulnerabilities that influence the perceived severity of food insecurity

All other controlled variables in the regression including household-level variables, such as household size, the number of young or elderly members, and household head characteristics (e.g., education, marital status), show no consistent or significant relationship with food insecurity outcomes. Similarly, variables related to structural poverty (e.g., impoverished roof or floor) exhibit small effects but are not statistically significant. These put together show that while structural and demographic variables may contribute to food security outcomes, their lack of statistical significance in this model suggests that the sex of the respondent is a strong predictors of food security in Burkina Faso using the FES and income source and income change-based CARI classifications.

6 Summary & conclusions

In this pilot study, we examined the effect of survey modality (Face-to-Face [F2F] vs. remote) on food security assessment in Burkina Faso. The study employed a survey experiment using a stratified random sample selected from three regions of the country with varying degrees of food insecurity and livelihoods. The experiment was implemented by randomly assigning sampled households to one of the two survey modalities using their baseline observable characteristics as control (Household baseline characteristics were collected during a household census phase that was conducted before the actual survey). The two samples (F2F vs. remote) were then tested for statistical comparability using their their baseline observable characteristics such as age of household head, household size, gender of household head, education of household head, main source of livelihood, dependency ratio, etc.

In total 2,882 households were interviewed (F2F and remote together), using identical food security assessment modules. Data collection was carried out over a four-week period, with both modalities implemented simultaneously by trained and experienced enumerators.

The data from this experiment provides important insights for conducting food security assessment using remote methods (phone survey). Below are some key highlights and talking points from the analysis of this pilot study.

- No significant impact of modality was found at the aggregate level for the food security prevalence reported across the two samples after controlling for respondent and household characteristics. This may indicate that difference arising from the results of the rCARI and CARI aggregation methodologies is driven by the individual indicators like the LCS and income change.
- Respondents in the remote sample generally tend to report better aggregated food security outcome (report better food security outcomes) than the F2F sample. Intra-household comparison of the two aggregation methodologies (CARI and rCARI) show that households would overstate their economic vulnerability using expenditure while under-reporting using the income choice and income change indicator. This result is consistent in the inter-household analysis of the responses for the two indicators.
- The Food Expenditure Share (FES) indicator is found to be stable across modalities at the aggregate level though this difference emerge when the comparison is done acros the 4-CARI FES categories.
- The analysis of the long- and short- LCSI, FES and FCS overall does not show any statistically significant difference. This means that by fine-tuning and increased training, the short modules could potentially replace the longer versions in F2F surveys to reduce survey fatigue both for the interviewee and interviewer. For example the shortened FES module piloted in the phone surveys produced results that were statistically comparable

to the standard (long module) FES module that was implemented in in the F2F survey.

- Using the income source and income change indicator may be effective for capturing short-term economic shocks and acute vulnerabilities that do not necessarily reflect the long term consumption patterns and therefore especially valuable for assessment during crises like natural disasters or economic downturns, where income changes directly impact food security.
- Further analysis is needed to understand the dynamics of marginally and moderately food-secure households, where the CARI classification are significantly different for the two aggregation methods, i.e., CARI_FES and CARI_Income. .
- In terms of the food security indicators, households tend to reports slightly higher food consumption (FCS) and better livelihood coping situations over the phone (remote modality) than when interviewed F2F.

The key communication points from the foregoing is that:

- **Survey Modality (F2F vs. Remote):** While challenges such as connectivity issues and lower response rates may not favor remote surveys, when properly implemented, remote surveys can produce food security assessments that are comparable to F2F surveys results.
- **Alignment of CARI and rCARI Methodologies:** CARI using the income-based indicator as a measure of economic vulnerability of the population tends to adopt a conservative approach in estimating acute food insecurity compared to the use of the FES regardless of the survey modality. Thus the main difference is not in the modality but in the income source and income change indicator which needs further study to align.

6.1 Recommendations

Based on the foregoing findings from the Burkina Faso study data, the following recommendations are deduced:

- The use of rCARI with the income source and income change indicator leads to under estimating the prevalence of AFI and thus in its current state if used results should be adjusted by on average 15 percentage points to align with CARI.
- More refinement of the income source and income change indicator should be pursued at a more global level to better categorize the different income sources into the 4-point CARI categories that can better reflect their economic vulnerability in remote surveys.
- Integrate and complement both Food Expenditure Share (FES) and the Income Choice & Income Change indicators in reporting CARI (CARI-FES and CARI-Inc): The use of both FES and the income source and income change indicators in tandem will capture both long-term trends and short-term vulnerabilities in food security assessments, particularly during crises.

- Address gender-specific vulnerabilities: Incorporate gender-sensitive approaches into food security assessments to account for reporting biases and gender-specific economic vulnerabilities, particularly in income-based indicators.
- Further testing, refining and optimizing the shortened modules for F2F Surveys particularly the expenditure module: Based on the findings that shorter modules yield similar results like the standard long ones, implementing shorter versions in face-to-face surveys to reduce respondent and enumerator fatigue (survey duration) as well as overall survey cost, without compromising data quality should be considered.
- Enhance enumerator training: Improving training for enumerators will to mitigate challenges associated with remote data collection, especially in populations where education is low.

7 Supplementary Analysis

The livelihood distribution between the two modalities showed a sizeable imbalance, with respondents reporting small scale farming as their main source of livelihood representing over 50% of the F2F participants and only 2% of the remote households. Given that small farmers are generally among the most food insecure and by extension the most vulnerable households, this imbalance is likely to cause confounding and complicate the ability to assess the role of modality through direct analysis. Additionally, there is a very sparse representation of non-agricultural livelihoods among both the F2F and remote participants. Therefore, to compare results for the various assignment groups, a subgroup of 1,397 households representing 589 (32%) of the F2F households and 808 (80%) of the remote households with primary income derived agricultural production has been analysed. This subset is analysed to compared response patterns for each component of CARI/rCARI to determine the extent to which modality contributes to differences in food security classifications.

The main results summarized from these analysis comparing the outcomes for the different indicators (FCS, LCS, rCSI, and FES) are available but have not been included in the current report.

7.1 Study limitations & implementation challenges

Though the study has been implemented with utmost regard for all theoretical and practical guidelines and protocols underpinning experimental survey designs, the study nonetheless recorded the low response for the remote sample in comparison to the F2F sample. This affected the overall final sample distribution at ADMIN5 level. However, it must be mentioned that in spite of the high iteration in the remote sample, the final sample still exhibited a high level of statistical comparability based on their observable characteristics as baseline.

Similarly, although extensive efforts were made to reduce and manage implementation challenges, some remained and affected the survey's successful implementation. These include:

1. Restricted access to the field because of security concerns.
2. Poor mobile phone network/infrastructure
    a. Poor network connectivity especially in the rural parts of the country affecting the response rate.
    b. Bad infrastructure such as bad roads and frequent power cuts affected affecting the inclusion of the hard to reach areas of the country in the study.
3. Weather disruptions/rainfall causing floods disrupting the road network to the selected villages. Also, most of the potential respondents were away for their farm work affecting overall response rate.

8 Next steps & further analysis...

1. Conduct further in-depth analysis (statistical diagnostics) of the data - exploring the association between FES and income indicators in particular.
2. Explore the actual categorization of the different income sources into the four rCARI income groups to identify trends across different pilot countries (this will feed into possible recalibration efforts of the income indicator to align more with the FES).
3. Prepare the draft manuscripts for journal publication.
4. Continue with the overall data collection in the remaining pilot countries (Iraq, Mauritania, & Guatemala)

9 References

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10 Annexes

10.1 Study areas selected for the pilot

Table 20: List of Selected ADMIN5 Areas & Number of HHs per Area

REGION	PROVINCE	COMMUNE	Village	Population	CH Class
PLATEAU CENTRAL	GANZOURGOU	MEGUET	Kougdoughin	240	IPC1
			Boulwando	460	IPC1
			Baghin	248	IPC1
			Nahoubé	270	IPC1
			Zemalga	623	IPC1
			Koulweogo	396	IPC1
			Tamasgo	397	IPC1
			Tibin	425	IPC1
			Vagma	255	IPC1
			Kabouda	360	IPC1
CENTRE NORD	SANMATENGA	KORSIMORO	Ilartenga	188	IPC3
			Nigui	193	IPC3
			Ouitenga-Poecin	210	IPC3
			Bendogo	249	IPC3
			Nimpoui	264	IPC3
			Tamboko	293	IPC3
			Noungou	408	IPC3
			Ipalla	469	IPC3
NORD	PASSORE	LA-TODIN	Minissia	355	IPC2
			Bissiga	329	IPC2
			Gogho	325	IPC2
			Baribsi	208	IPC2
			Kingria	592	IPC2
			Sissamba	254	IPC2
			Kollo	396	IPC2
			Loungo	852	IPC2
			Kalamtogo	259	IPC2
NORD	YATENGA	OULA	Bougoure	596	IPC3
			Bourbo	235	IPC3
			Pelle	379	IPC3
			Omsom	412	IPC3
			Oula	313	IPC3
			Lougouri	647	IPC3
			Zanna Mossi	236	IPC3
			Sonh	229	IPC3
			Tilli	220	IPC3
			Kayegue Mossi	248	IPC3

10.2 Randomization balance test

Table 21: Balance Table for Randomized Survey Modality Assignment: Face-to-Face vs. Remote

Variable	F2F	Total Remote	Diff
List respondent age	45.69	45.18	-0.51
	(16.80)	(16.21)	(0.31)
Household head is the respondent	0.76	0.78	0.01
	(0.42)	(0.42)	(0.26)
Female Household Head	0.20	0.20	-0.00
	(0.40)	(0.40)	(0.98)
Married Household Head	0.97	0.98	0.01
	(0.16)	(0.14)	(0.26)
Household head years of education	1.38	1.34	-0.04
	(2.89)	(2.85)	(0.65)
Displaced household	0.03	0.04	0.01*
	(0.17)	(0.20)	(0.05)
Household size	7.14	7.18	0.04
	(4.34)	(4.31)	(0.76)
Number of household member younger than 15	3.15	3.18	0.04
	(2.42)	(2.43)	(0.64)
Number of household member older than 60	0.59	0.58	-0.01
	(1.05)	(0.98)	(0.77)
Has household member with disability/chronic disease	0.18	0.16	-0.02
	(0.38)	(0.36)	(0.05)
Number of rooms	1.69	1.65	-0.04
	(1.15)	(1.08)	(0.25)
Household Crowding Index	4.94	5.06	0.12
	(3.71)	(3.74)	(0.31)
Household has impoverished roof	0.01	0.01	-0.00
	(0.12)	(0.11)	(0.51)
Household has impoverished floor	0.16	0.16	0.00
	(0.36)	(0.37)	(0.71)
Observations	2,131	2,125	4,256

Note: Robust standard deviations in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Table 22: Balance Table for Randomized Questionnaire Length Assignment in F2F Survey: Long vs. Short

		F2F	
Variable	Long	Short	Diff
List respondent age	46.29	45.09	-1.20
	(16.88)	(16.71)	(0.10)
Household head is the respondent	0.76	0.77	0.00
	(0.43)	(0.42)	(0.93)
Female Household Head	0.19	0.21	0.02
	(0.39)	(0.41)	(0.38)
Married Household Head	0.98	0.97	-0.02*
	(0.13)	(0.18)	(0.03)
Household head years of education	1.31	1.46	0.15
	(2.82)	(2.95)	(0.24)
Displaced household	0.03	0.03	-0.00
	(0.18)	(0.17)	(0.90)
Household size	7.11	7.16	0.04
	(4.31)	(4.36)	(0.81)
Number of household member younger than 15	3.15	3.14	-0.01
	(2.43)	(2.42)	(0.89)
Number of household member older than 60	0.60	0.58	-0.03
	(1.01)	(1.10)	(0.55)
Has household member with disability/chronic disease	0.17	0.18	0.01
	(0.38)	(0.39)	(0.39)
Number of rooms	1.66	1.72	0.06
	(0.98)	(1.30)	(0.25)
Household Crowding Index	4.94	4.94	0.00
	(3.58)	(3.84)	(0.99)
Household has impoverished roof	0.02	0.01	-0.00
	(0.12)	(0.12)	(0.86)
Household has impoverished floor	0.16	0.15	-0.01
	(0.37)	(0.36)	(0.44)
Observations	1,066	1,065	2,131

Note: Robust standard deviations in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Table 23: Balance Table for Randomized Questionnaire Length Assignment in Remote Survey: Long vs. Short

		Remote	
Variable	Long	Short	Diff
List respondent age	44.67	45.69	1.01
	(16.16)	(16.25)	(0.15)
Household head is the respondent	0.76	0.80	0.04*
	(0.43)	(0.40)	(0.04)
Female Household Head	0.19	0.21	0.02
	(0.39)	(0.40)	(0.37)
Married Household Head	0.98	0.98	0.00
	(0.15)	(0.13)	(0.45)
Household head years of education	1.41	1.27	-0.14
	(2.94)	(2.75)	(0.27)
Displaced household	0.04	0.04	0.00
	(0.20)	(0.20)	(0.90)
Household size	7.06	7.29	0.22
	(4.33)	(4.30)	(0.23)
Number of household member younger than 15	3.15	3.21	0.06
	(2.44)	(2.42)	(0.58)
Number of household member older than 60	0.55	0.61	0.05
	(0.89)	(1.06)	(0.21)
Has household member with disability/chronic disease	0.15	0.16	0.01
	(0.36)	(0.37)	(0.74)
Number of rooms	1.64	1.66	0.02
	(1.11)	(1.05)	(0.69)
Household Crowding Index	5.00	5.11	0.11
	(3.69)	(3.79)	(0.50)
Household has impoverished roof	0.02	0.01	-0.01
	(0.12)	(0.10)	(0.24)
Household has impoverished floor	0.18	0.14	-0.03*
	(0.38)	(0.35)	(0.03)
Observations	1,064	1,061	2,125

Note: Robust standard deviations in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

10.3 F2F Attrition Tables

Table 24: Balance Table for Randomized Questionnaire Length Assignment in F2F Survey: Completed vs. Attrition

Variable	Completed	F2F Attrition	Diff
List respondent age	45.48	47.12	1.64
	(16.79)	(16.82)	(0.14)
Household head is the respondent	0.77	0.74	-0.03
	(0.42)	(0.44)	(0.31)
Female Household Head	0.20	0.20	-0.00
	(0.40)	(0.40)	(0.96)
Married Household Head	0.98	0.97	-0.01
	(0.15)	(0.18)	(0.42)
Household head years of education	1.42	1.12	-0.31
	(2.92)	(2.64)	(0.08)
Displaced household	0.03	0.04	0.01
	(0.17)	(0.20)	(0.39)
Household size	7.11	7.33	0.22
	(4.36)	(4.18)	(0.41)
Number of household member younger than 15	3.14	3.21	0.08
	(2.42)	(2.43)	(0.63)
Number of household member older than 60	0.58	0.65	0.07
	(1.07)	(0.97)	(0.29)
Has household member with disability/chronic disease	0.18	0.17	-0.01
	(0.38)	(0.37)	(0.64)
Number of rooms	1.69	1.64	-0.05
	(1.17)	(0.98)	(0.43)
Household Crowding Index	4.92	5.13	0.21
	(3.77)	(3.30)	(0.33)
Household has impoverished roof	0.02	0.01	-0.00
	(0.12)	(0.11)	(0.58)
Household has impoverished floor	0.15	0.19	0.04
	(0.36)	(0.39)	(0.13)
Observations	1,863	268	2,131

Note: Robust standard deviations in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

10.4 Remote Attrition

Table 25: Balance Table for Randomized Questionnaire Length Assignment in Remote Survey: Completed vs. Attrition

Variable	Completed	Remote Attrition	Diff
List respondent age	45.65	44.75	-0.90
	(16.31)	(16.11)	(0.20)
Household head is the respondent	0.82	0.75	-0.07***
	(0.39)	(0.44)	(0.00)
Female Household Head	0.19	0.21	0.02
	(0.39)	(0.41)	(0.18)
Married Household Head	0.99	0.97	-0.01
	(0.12)	(0.16)	(0.08)
Household head years of education	1.42	1.27	-0.15
	(2.94)	(2.76)	(0.23)
Displaced household	0.04	0.04	0.00
	(0.20)	(0.21)	(0.73)
Household size	7.42	6.95	-0.46*
	(4.36)	(4.26)	(0.01)
Number of household member younger than 15	3.26	3.11	-0.15
	(2.39)	(2.46)	(0.16)
Number of household member older than 60	0.64	0.53	-0.11*
	(1.04)	(0.91)	(0.01)
Has household member with disability/chronic disease	0.17	0.15	-0.02
	(0.37)	(0.35)	(0.22)
Number of rooms	1.68	1.62	-0.07
	(1.20)	(0.96)	(0.16)
Household Crowding Index	5.21	4.92	-0.30
	(4.02)	(3.46)	(0.07)
Household has impoverished roof	0.01	0.01	-0.00
	(0.12)	(0.10)	(0.32)
Household has impoverished floor	0.17	0.15	-0.02
	(0.38)	(0.36)	(0.22)
Observations	1,019	1,106	2,125

Note: Robust standard deviations in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

10.5 Supporting documents

10.5.1 CARI resources

- CARI methodology Consolidated Approach for Reporting Indicators of Food Security (CARI) Guidelines - Third edition
- FCS
- rCSI
- LCSI
- FES